Wiqaya: Wildfire Susceptibility Mapping and Alerts for the Forests of Northern Morocco

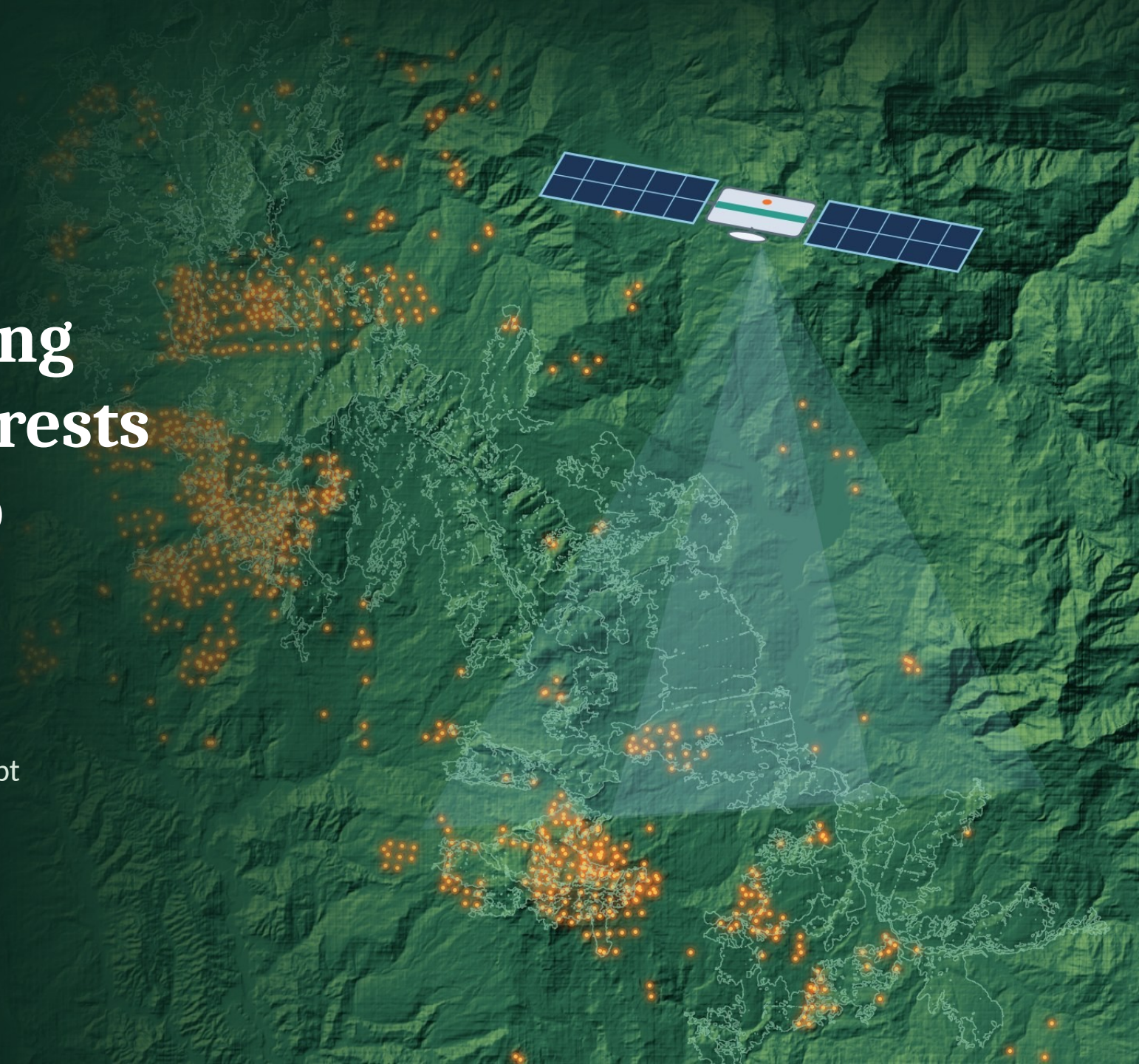
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Team Wiqaya — Morocco

Arab Youth Space Hackathon 2026 · Proof of Concept

Theme: Climate & Disaster Resilience

October 2026 · github.com/SanaeBelfrouh/wiqayagit



The western Rif burns almost every summer, and warnings arrive after ignition

- **Recent large fires:** Bouhachem 2022 (> 7,500 ha), Larache 2022 (> 10,500 ha), Derdara 2021 and Bouhachem 2020 (~1,100 ha each), two fires in August 2025
- **Drivers:** hot, dry chergui winds, steep slopes, villages (douars) inside the forest
- **Gap:** danger bulletins are province-scale and satellite detections come after ignition; there is no daily, cell-level view of risk
- **Why satellite data:** NASA FIRMS gives the only consistent 20-year fire record of these remote mountains (MODIS since 2005, VIIRS 375 m since 2017)

10,600
people living inside the forest

264,000
people within 10 km of it

445
FIRMS detections inside the forest since 2005

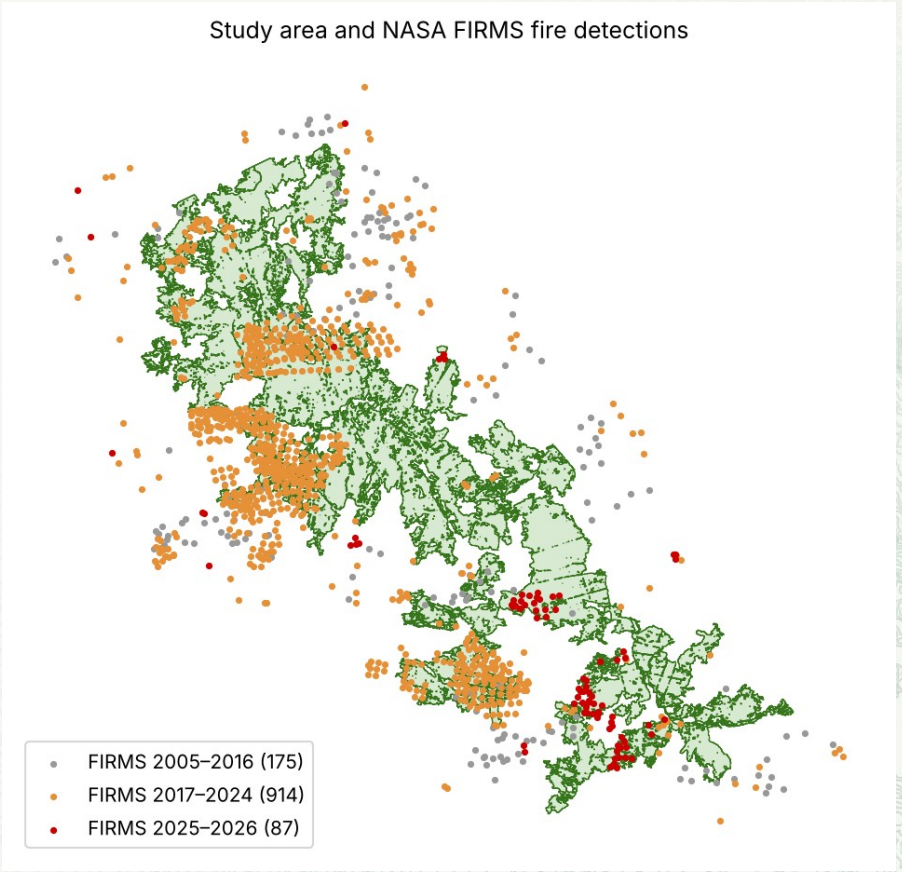


Fig. 1. Study area (25,900 ha) and NASA FIRMS detections by period. Population: WorldPop 2017.

Business use case: where to patrol today, and whether to warn residents

End users	Forest managers (national forestry agency, ANEF), Civil Protection, provinces and communes; residents of the villages in and around the forest
Decision supported	Each summer morning: where to send patrols and first-intervention teams, and whether the day is dangerous enough to warn residents and restrict burning
Current practice	Province-scale danger bulletins, field knowledge, and fire detections (FIRMS, EFFIS) that arrive after ignition; no alerts in Arabic and French
Contribution of Wiqaya	250 m risk map with alert zones and a per-cell explanation, daily weather danger index, 20-year fire history, SMS / email alerts in three languages, open GeoJSON API
Business model	Subscription per protected area for institutions; free citizen alerts funded by these contracts or CSR partners; open data, so no imagery cost

Data: open Earth-observation and ancillary datasets

Dataset	Product and provider	Period	Use
Active fires	NASA FIRMS: MODIS C6.1 and VIIRS S-NPP 375 m	2005–2026 (1,176 points)	Labels, fire history
Terrain	FABDEM V1-2, 30 m (Univ. of Bristol; Copernicus GLO-30)	2023 release	Elevation, slope, aspect
Population	WorldPop mar_ppp_2017 (CC BY 4.0)	2017	Exposure per cell
Weather	Open-Meteo: ERA5 archive and forecast (CC BY 4.0)	2017–today	Weather danger index
Optical imagery *	Sentinel-2 L2A, Landsat C2 L2 (Planetary Computer)	2005–2025	NDVI, NDMI, NBR, dNBR
Land cover, OSM *	2021 classification; OpenStreetMap 2022	2021–2022	Fuel, human pressure

* *Research notebook only.* **Hyperspectral data is not used in this proof of concept;** EnMAP or Arab Satellite 813 scenes are planned to map live fuel moisture and vegetation species.

Approach: from satellite detections to alert zones

1. Inputs
FIRMS 2005–2026, DEM, WorldPop, forest boundaries

2. Grid and features
6,071 cells of 250 m; elevation, slope, aspect, population

3. Fire history
FIRMS within 375 m; years since last fire (no leakage)

4. Model selection
Logistic regression vs random forest, leave-one-year-out 2017–2024

5. Fuel rule
Risk 0 if burned < 1 year ago; × 0.5 if 1–5 years

6. Blind test
Retrain on 2017–2024; score the 2025 fires

7. Risk map
5 KMeans classes; alert = riskiest 20 % of cells

8. Weather danger
Max T, VPD, wind, min RH ranked vs 2017–2024 summers

Reproducible: notebooks/02_main_analysis.ipynb runs end to end in about 5 minutes from the committed sample inputs (fixed seeds, relative paths).

Risk classes on 1 August 2025: the 2025 fires fell outside the alert zones

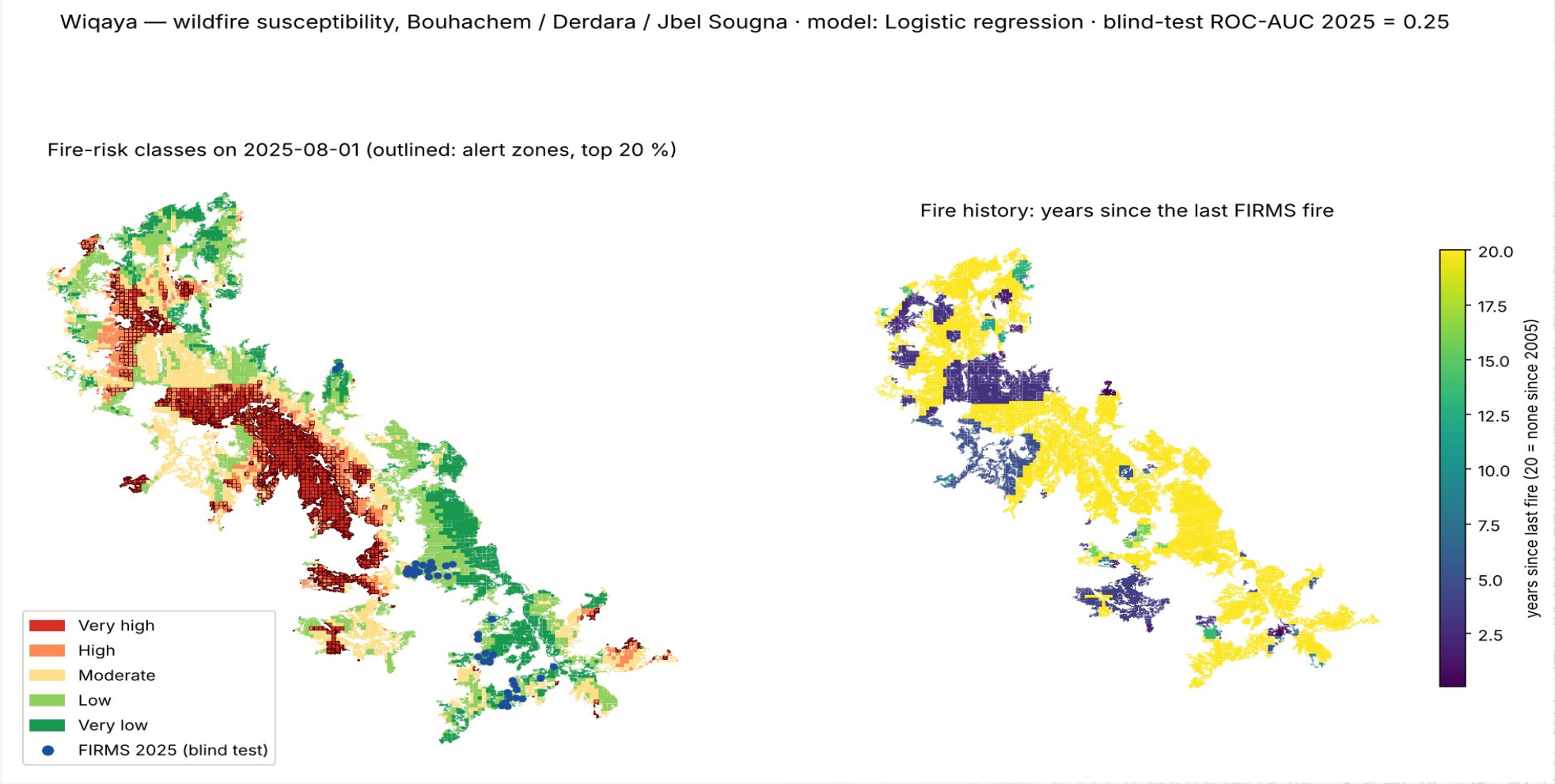


Fig. 2. Left: five risk classes; outlined cells are alert zones (riskiest 20 %); blue points are the 2025 FIRMS fires (blind test). Right: years since the last FIRMS fire (20 = none since 2005). Notebook output results/example_output.png.

Population, terrain and fire history drive the selected model



Fig. 3. Absolute standardised coefficients of the logistic regression.

- **Selected model:** L2 logistic regression (C = 0.5, balanced classes)
- **Strongest predictors:** population, elevation, east-facing aspect, then fire history (years since the last fire, burned in the last 5 years)
- **Weak predictor:** slope
- **Missing drivers:** vegetation dryness and human access, available only in the research notebook

The same model is delivered to users through a web platform

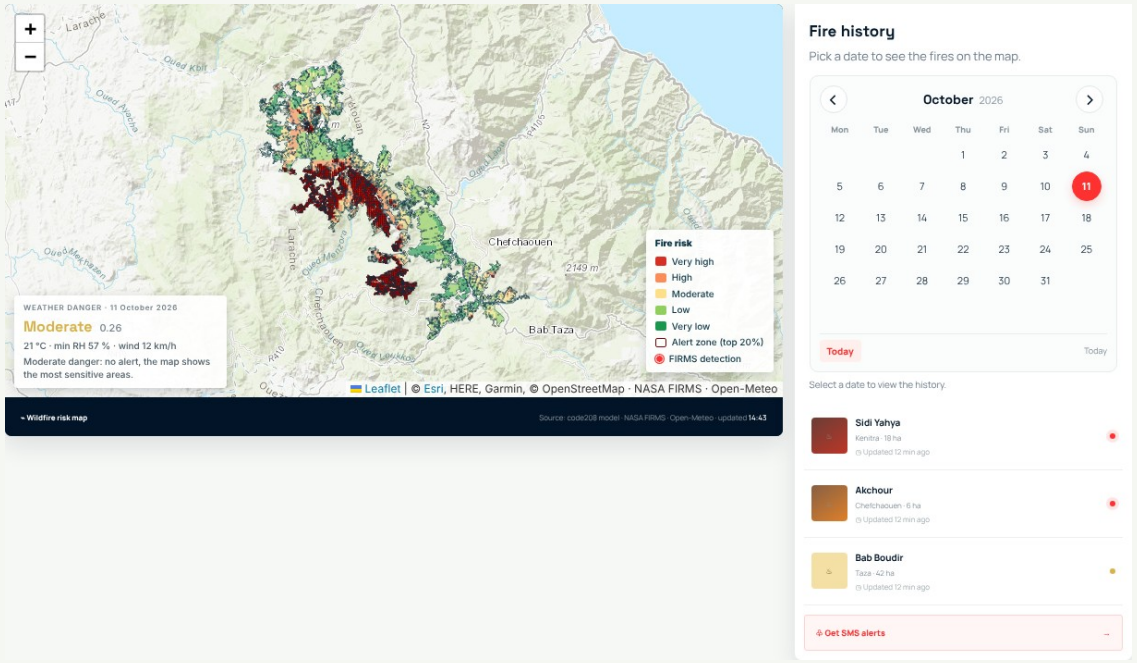


Fig. 4. Daily risk map with the weather danger of the day and the five-class legend.

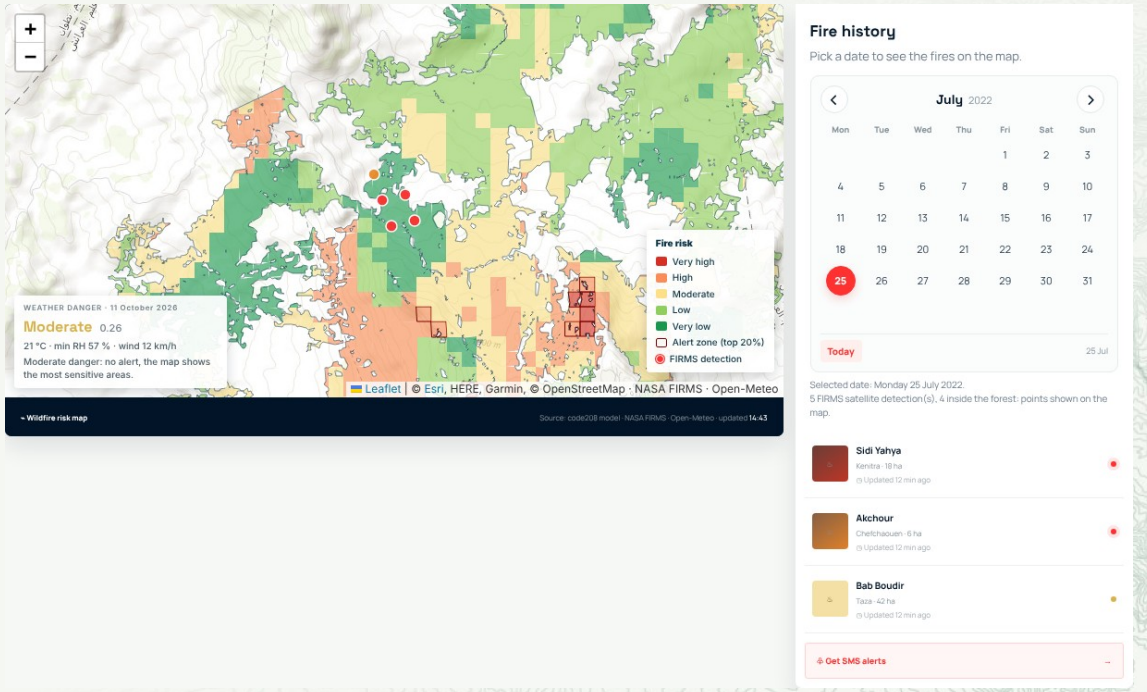


Fig. 5. Fire history: 25 July 2022, start of the Bouhachem fire; FIRMS shows 5 detections, 4 inside the forest.

Access for residents: Arabic, French and English, on any phone

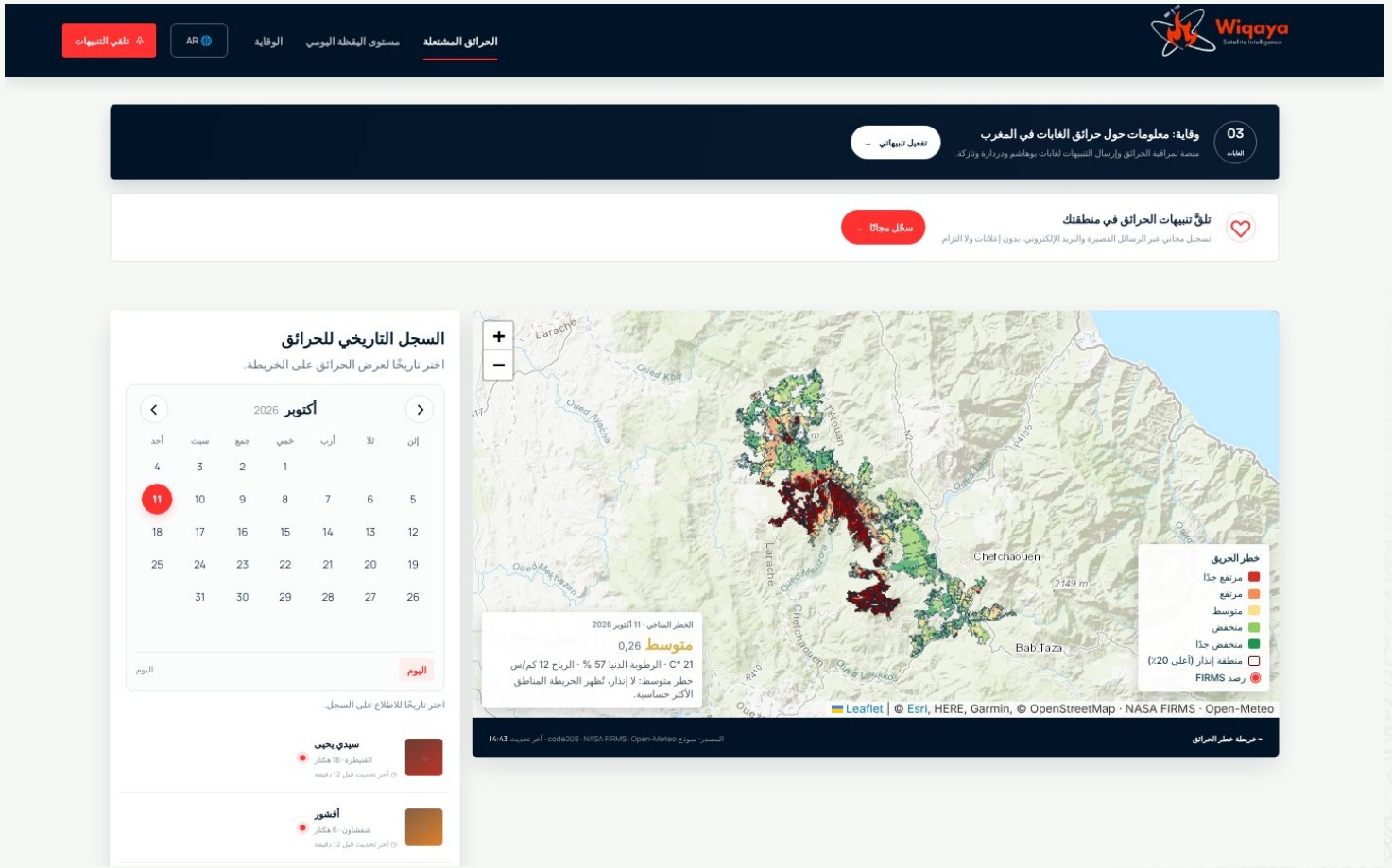


Fig. 6. Right-to-left Arabic interface.

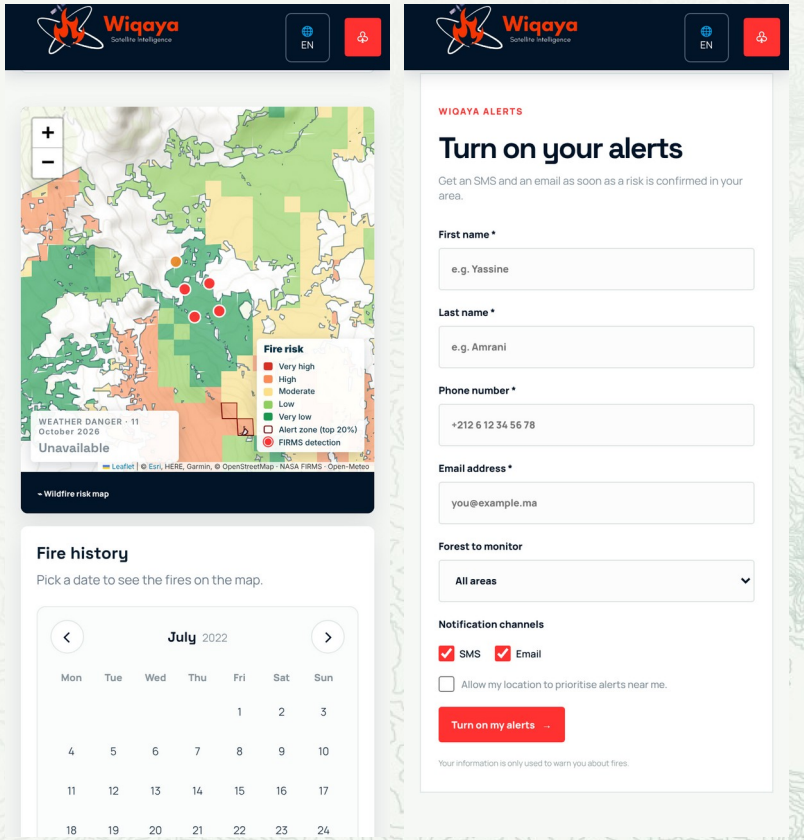


Fig. 7. Fire history and SMS / email alert sign-up on a phone.

Validation: informative on large-fire years, but the 2025 blind test fails

Test	Result
Leave-one-year-out 2017–2024, logistic regression (selected)	Mean ROC-AUC per year 0.62; global 0.65; PR lift 1.64
Leave-one-year-out 2017–2024, random forest	Mean ROC-AUC per year 0.56; global 0.60
ROC-AUC per held-out year	2018: 0.99 · 2019: 0.88 · 2020: 0.80 · 2021: 0.68 · 2023: 0.23 · 2024: 0.06
Blind test, 2025 fires (39 cells)	**ROC-AUC 0.25; no fire cell in the alert zones**

- **Works** on the large fire years 2018–2021
- **Fails** on 2023–2025: the 2025 fires started in cells rated low risk
- **Cause:** predictors describe exposure, terrain and history, not vegetation dryness or human access
- **Few positives:** 345 fire cell-years; per-year scores are unstable
- **No leakage:** history strictly before each date; 2025 never used for training or selection

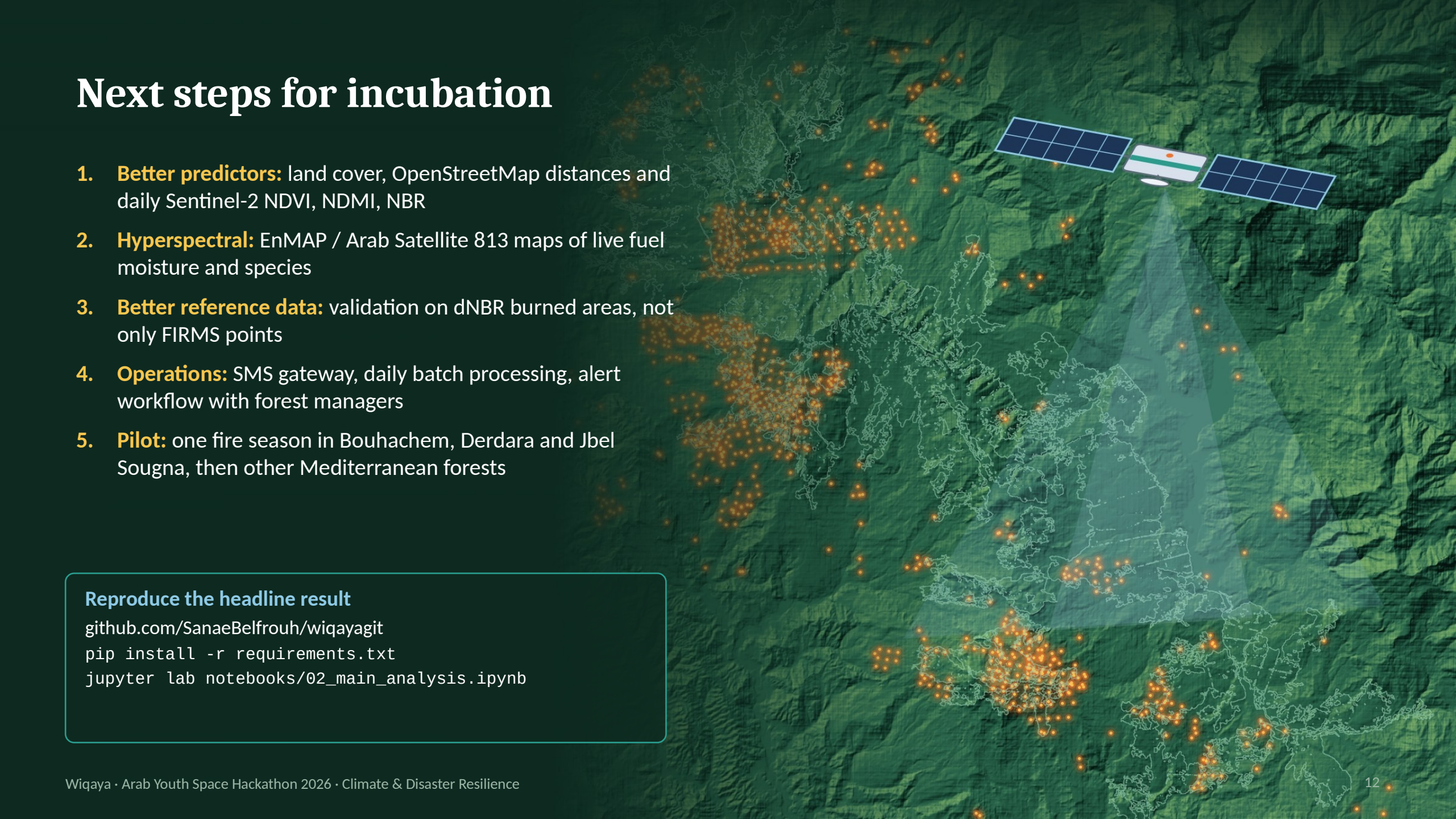
Expected impact and strategic alignment

Goal	Contribution
SDG 13 Climate action (13.1)	Resilience to heatwave- and wind-driven fire seasons
SDG 15 Life on land (15.1, 15.2)	Patrols and firebreak work targeted at the riskiest cells
SDG 11 Sustainable communities (11.5)	Early warning for ~10,600 people living inside the forest

- **Morocco:** supports the strategy Forêts du Maroc 2020–2030, ANEF and Civil Protection
- **Arab region:** open data only; replicable in Mediterranean forests of Algeria, Tunisia, Lebanon and Syria
- **Inclusion:** Arabic-first interface; SMS for residents without smartphones

Next steps for incubation

1. **Better predictors:** land cover, OpenStreetMap distances and daily Sentinel-2 NDVI, NDMI, NBR
2. **Hyperspectral:** EnMAP / Arab Satellite 813 maps of live fuel moisture and species
3. **Better reference data:** validation on dNBR burned areas, not only FIRMS points
4. **Operations:** SMS gateway, daily batch processing, alert workflow with forest managers
5. **Pilot:** one fire season in Bouhachem, Derdara and Jbel Sougna, then other Mediterranean forests



Reproduce the headline result

github.com/SanaeBelfrouh/wiqayagit

```
pip install -r requirements.txt
jupyter lab notebooks/02_main_analysis.ipynb
```

References and data sources

Assali, F. et al. (2016). Forest-fire risk mapping in the Chefchaouen-Ouazzane region, Morocco (topoclimate index).

Hawker, L. et al. (2022). A 30 m global map of elevation with forests and buildings removed (FABDEM). Environ. Res. Lett., 17, 024016.

Hersbach, H. et al. (2020). The ERA5 global reanalysis. Q. J. R. Meteorol. Soc., 146, 1999–2049.

NASA FIRMS. MODIS Collection 6.1 and VIIRS S-NPP 375 m active fire detections. https://earthdata.nasa.gov/firms

WorldPop (2018). Morocco 2017 population counts (mar_ppp_2017). University of Southampton. CC BY 4.0.

Zippenfeld, P. Open-Meteo.com Weather API. CC BY 4.0. https://open-meteo.com

Copernicus Sentinel-2 L2A data (2015–2025) and USGS Landsat Collection 2 Level-2, accessed via Microsoft Planetary Computer.

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Software: GeoPandas, Rasterio, rasterstats, scikit-learn, FastAPI, Leaflet. Code released under the MIT licence.